

37 STREET SMITH PUBLICATION

ASTOUNDING

REG. U. S. PAT. OFF.

Science Fiction

DECEMBER, 1945

25 CENTS

BEGGARS
IN
VELVET

BY LEWIS PADGETT

KNOW THIS EMBLEM ...
THE DISCHARGED
VETERAN WEARS
THIS AS HE WORE
HIS UNIFORM ...
WITH HONOR!



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Trouble Times Two

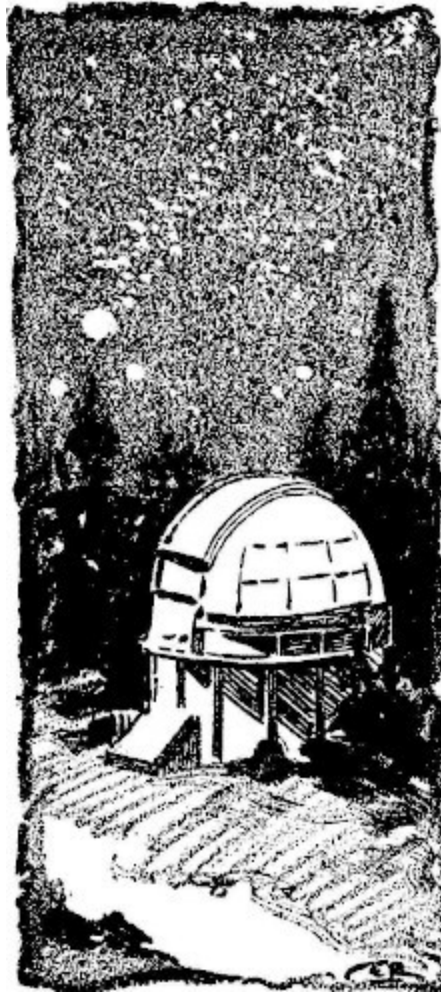
By GEORGE O. SMITH

Illustrated by Raymond

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Thomas Lionel Ph.D., M.M. bounded out of bed with a cheerful bit of off-tune song. He glanced at the calendar and then the clock and he grinned because life was just too good to be true.

Everything was according to plan. He'd won his first battle. Up to now it had been touch and go; at last he had established his right to co-occupy the mind along with the engineer. No longer could the engineer claim that he was an expensive detriment. He had forced the engineer into agreeing that his offering, though not directly productive, was a causative factor in the development of success. Then to top it all, he retained enough technology to be a necessary item. He must be permitted to remain if only for a source of information.



The engineer's trap had been excellent. But the trap had turned and caught the engineer. Those reams of data on the poltergeist effect had been the basis for an entirely new science that only a real physicist could appreciate—and no engineer could hope to thread his way through them without a research physicist's assistance.

He stood over the chessboard in the living room for a few minutes. The engineer was not making any great moves. Therefore the physicist thought that he might best consolidate his position. He castled to the queen's side, burying his king behind a bulwark of defenses that would defy a master chess player to penetrate in less than ten or fifteen moves.

During breakfast, he perused a thin volume of recent publication. He did not entirely agree with the theories presented; after all, the book had been written for the express purpose of getting reader's viewpoints and Thomas knew it. In fact, the book was not too interesting to Thomas but he knew that the engineer would fume, fret, and howl at the idea of having a well-thumbed volume of "*Theory of Multi-Resonant Wave Guides*" in the library.

Thomas wouldn't look at the engineer's volume, laying on the table opposite. It was too un-physical. It was un-erudite. It was "*Basic Theory in Micro-Wave Transmission*" and the edges of the pages were loaded with application formulas, diagrams, and working sketches.

He was near the end of breakfast when the glint of reflected sunshine arrowed through the window and caught his eye. He looked, and wondered who was landing on his lawn in a helicopter.

Then he did a double take.

"Helicopter" stemmed from Greek, the "helix" or screw plus the "opter" a machine. This contrivance did not. It was not operated with air screws.

It looked like a three-wheeled coupé. It looked like the industrial designer's dream of the Plan For Tomorrow, excepting those three wheels. The Plan For Tomorrow should, by all rights, have four wheels. And, if the thing is going to fly, it should have some sort of overhead vanes, or wings, or engines, or jets, or even a skyhook. But there it was, coming down as light as a feather to make a neat landing on the back lawn.

By the time the door was open, and the passenger stepped to the ground, Thomas was standing before the little sky car, looking somewhat dazed at the name:

POLTERGEIST

"Like a dream," said the driver of the sky car.

"It should," said Thomas, covering his ignorance with monosyllabic agreement.

"Handles well, too. I think we could stand a bit more positivity of control, though."

"I'll look into it."

"I wish you would. We've got the jump on the whole world with this. We'd like to keep it. But the thing doesn't answer to the wheel too solidly."

"Uh-huh."

"The chief engineer said, 'Jim, take that crate over to Lionel and see if he will beef up the control force a bit.' So here I am."

"O.K., Jim," said Thomas, offering a prayer for the name that had been given unwittingly. The engineer must have been a busy boy! "How are you going to get back?"

Jim looked up into the sky. "Jerry is following in the pilot model. He'll pick me up and we'll go on back thataway."

Jim nodded skyward, and Thomas looked at the growing speck that must have been the pilot model.

Thomas forgot about the pilot model. What he wanted to know was the whereabouts of the five tons of equipment that had been an integral part of this idea. He looked at the model. He wondered whether the engineer had installed the whole thing, stepping up the power and using the main part of the power to support the equipment. That did not seem possible. Any failure would cause the little sky car to collapse of its own dead weight. Besides there was not enough room in the little crate to pack all that equipment-tonnage.

The engineer had achieved the impossible. He had done away with the main part while retaining the effect.

The pilot model landed. It was not the finished job of the prototype. The cabin was squarely functional and the landing wheels were not faired into the hull. The rear end, instead of tapering gently into a narrow paraboloid of revolution, was a truncated four-sided pyramid.

Jerry did not emerge. He merely tossed the door open and shouted: "Come on—we ain't got all day!"

Thomas nodded. "I'll call you when I get it fixed."

Call who? the physicist wondered, and then forgot about it. He wanted desperately to dig into the sky car. He wanted to find out where the engineer had packed five tons of equipment. He wanted to see what made the wheels

go around. No doubt the thing could be returned to its owners without calling in the police. The thing was probably recorded in the precisely kept engineering notebook of the physicist's alter ego.

The pilot model was not completely out of sight before Thomas had the power cowls off, and the whole model stripped of its servicing doors. They had done an excellent job of design; the sky car without its servicing panels was but a skeleton frame, with every line, every connection, and every control rod open for easy servicing.

And it was then and there that the physicist understood what the engineer had been doing.

Instead of the low-voltage high-current supply lines, with their attendant heavy busbars, thin pipes ran about the sky car. Seamless aluminum tubing carried the energizing current. Or, rather the space inside of the tubing carried it. At the generator end, a ten megawatt microwave generator supplied high power at ultra high frequency. At the terminus, rectifiers brought the ultra high frequency down to direct current for operation of the force-field generators.

Thomas nodded. It was not the final tenth of one percent job. It was not direct current. The diagravitic force was not constant. It operated only seven tenths of the time, and was turned off and on fifteen or twenty million million times per second. Nothing short of high-definition test equipment would ever tell the difference, however.

Gone were the massive electromagnetic deflection field coils. In their place was a set of seventy kilovolt electrostatic plates.

Missing entirely were the variable-speed motor generators. In their place was a simple crystalline formation under permanent magnetic stress. "Artificial radio-activated crystals," muttered Thomas. "Good for a couple of years."

But the feed lines. *The feed lines.* The current carrying ability of space itself—not the metallic conductor—did the trick. Using the ultra high frequency bands, the busbars had been replaced with cylindrical wave guides. The

depth of penetration was measured in microns at those frequencies—and as long as the guides were properly designed, they offered little loss in power. The current went down the wave guides by virtue of the magnetic fields created throughout the guides—magnetic fields generated in the space inside of the tubular guides.

The generator itself was one of the new crystal microwave generators and the rectifiers at the receiving end were of the same ilk.

And the five tons of equipment had vanished in a puff of tubular guides, electrostatic plates, and intermittent operation.

Thomas hit a snag for a moment. The engineer had answered his challenge. So he'd come up with the answer to the five-ton-per-fifty-pound answer—and had gone further. Thomas knew that there was no apparent limit to the maximum power or lift. It merely set a fifty pound minimum—actually it was 49.87 pounds by measurement—under which limit no amount of tinkering would produce the effect.

He smiled. There must be something beyond. After all, small stones moved quietly in natural poltergeist manifestation; they would be able to reproduce that eventually. But for now, the engineer was willing to accept the limitation whereas the physicist would not.

He knew now. And he'd leave the sky car until the engineer returned. Let *him* beef up the control force. It was *his* baby.

Thomas put the panels back on the sky car and stood off to admire it. It was a neat job, just what the public wanted. The urge to get in and drive was a most compelling one, and Thomas succumbed. He sat for a moment, inspecting the dashboard until he had the pattern well set. Then he snapped on the power, took the wheel and pulled back gently. The sky car lifted its nose slightly, and as Thomas pressed the foot pedal, it took off on a side-line straight into the sky. He leveled off at a thousand feet and he did some scurrying back and forth in midair. It did handle a little sloppy but not enough to make the physicist uncomfortable. Yet it wouldn't stand any hedge-hopping or bridge-undercutting without a prayer on the part of the driver. Butter the controls a bit and you could thread a needle with it on the first try.

Yes, the engineer had done it again—all of which made Thomas chuckle. A bit more of this and the engineer would have such an income that he'd no longer worry himself into engineering. Then—

Thomas turned the sky car and drove across the city toward Dr. Hamilton's place. He landed on the psychiatrist's lawn and startled the doctor out of a week's growth.



"I've won," he told the doctor.

"Good," laughed Hamilton. "Mind if I ask which you are today—and how do you know you've won?"

"I'm Thomas Lionel, Ph.D. And the engineer has worked himself out of a job."

"Interesting. But how?"

"He dropped me a mess of cockeyed data, remember? Well, I unraveled it into a most interesting field of science. From it I handed him a slab full of theories and experiments that are just inefficient enough to make him fume. He's come up with several things that make money in vatfuls."

"That, I know and understand. Go on."

"Remember, I am his ideal personality, I am a physicist, a type of person he has always wanted to be. He couldn't be a physicist because of financial reasons and so he went into the engineering field to bolster up his bank account. That was eminently practical. But now that the worry about the bankroll is over, he can turn to theoretical physics and physical research. That's me—and I've won!"

"Suppose he, himself, takes the gradual retreat from engineering into physical research?"

"Um—I don't think he's capable of it. He's been too well conditioned."

"Might well be," admitted the doctor. "Well, as I said before, I'm just a referee. Both of you are well adjusted and good, worthy additions to society. Either one of you that wins will be a credit to civilization."

"You're a great help," laughed Thomas. "But I don't mind. This is my round, and it's my game. He's licked himself."

"I'll tell him that when I see him," said Dr. Hamilton. "But there is one thing that I must know. I want to know what makes that little tungsten box work."

"I cast the tungsten in—"

"I don't care how you made it," said Hamilton flatly, "unless it has a definite bearing on how it works."

"I made it of tungsten because the engineer would rip it apart if it weren't too tough," grinned Thomas. "Being of tungsten it doesn't matter how it

works excepting it would have been more efficient if I'd made it of silver."

"Look, Thomas, stay on the subject. I want to know what's with the works."

Lionel laughed. "What's so important?"

"Look, man, I'm a psychiatrist. The functioning of the human mind is my baby. Or," he added bitterly, "it should be. But, darn it, all we can do is to surmise, theorize, hope and pray. We don't know what makes schizophrenics, or manic-depressives or any of the other mental quirks. We aren't even certain why some people are well liked while others, of almost identical get-together are heartily disliked. But you've come up with a little dingus that causes a switch-over from one personality to another merely by pushing a button. Find out why and we psychiatrists may some day get to first base in psychoanalysis."

"Um—I suppose a real pathophone would be a help."

"Pathophone is a good word," smiled the psychiatrist, "but to dig into a warped mind without having the erroneous impressions and false evaluation clouding the only entry ... we'd be able to clear up almost any mental condition. Now, how does it work?"

"I am not prepared to say. I was seeking experimental data on the 'epicenter' of the poltergeist phenomena—the poltergeist usually manifests in the vicinity of or because of some central influence—usually a person who is unaware of his potentiality. At any rate, I was setting up a series of local magnetic and electrostatic fields and then trying the micro-microwave spectrum for response. I was running up through the region between long heat radiation and micro-micro radio waves when—blooey!—I was the engineer. I switched back eventually and consolidated my findings into that little tungsten box."

"I want the dope on it."

"I'll give it to you," nodded Thomas. "As soon as I make some final measurements and consolidate my data."

"Fine. Mind telling me what causes the poltergeist?"

"As best I can. The present concept of space is that space itself is under internal strain. Force vectors in cancellation prevail, resulting in a stable continuum. Space is warped by electrostatic effects, magnetic effects, and gravitic effects. These local effects do not create a discontinuity in the space strain, and therefore no eruption takes place. Now enters the epicenter. Radiation from his mind or brain in thinking goes out and starts a very minor sympathetic oscillation in the warps and strains of space. If these strains are in the right vectorial situation, the minor oscillation builds up the response amplitude—"

"That doesn't make sense," objected the doctor. "Mental radiation must be weak. How can it induce high power?"

"It can't. But if you know radio at all, you'll recall that a high 'Q' circuit will develop very high voltages across the terminals with a very small driving voltage. Well, this is analogous to the epicenter effect. The epicenter wave causes instability in the space strains because the brain wave is not a natural phenomena of space. Then—like two sticks end to end under compression, it takes very little sidewise thrust to make the compression-force collapse, forcing the sticks out at right angles. Follow?"

"But where did this energy or force come from?" puzzled Hamilton. "Isn't that a violation of the Law of Conservation of Energy?"

"Not at all. The law is still valid. It does state that you cannot get more out of anything than is put into it. The guesswork comes in deciding how the energy got there. Coal, for instance, is just a black stone. It has potential energy which was put into it by the eons of solar energy shining on the carboniferous forests. A stone has potential energy for falling. Where did it get it? It may have been carried up the hill; it may have been dropped from space—put out there by the cosmic eruption that caused Creation. Or it may have been on the edge of a gully and the potential drop made by the stream eroding the ground out from under it."

"How about atomic power?"

"You mean, how did the power get locked in the atom?"

"Yes."

"The power in the atom was put there by the universe's atom factories. Sol, and the other suns," explained Hamilton.

"But where did the earth—?"

"Creation," murmured Thomas. "Who knows? I don't. Every time somebody comes up with a perfect answer, someone else comes up with perfect data that proves that the answer couldn't be *anything* that anybody has ever used before.

"The atom factory is the Solar Phoenix. You start with hydrogen and carbon. The solar heat is such that they combine atomically to an unstable isotope of nitrogen which immediately becomes a stable isotope of nitrogen. More hydrogen gets in, making it unstable oxygen and so forth. Oxygen breaks down, releasing energy, helium, and, what do you know, carbon again, which begins to take on hydrogen again, and here we go again. But the thing is uncontrolled hell on wheels. Things go wrong due to the variances of pressure and temperature, and the oxygen doesn't always break down into helium and carbon. It takes offshoots and sidetracks. It'll add hydrogen and become fluorine, for instance, which then adds more and becomes something else, some of which trails off like the branches of a tree and do not break down into recurrent reactions. Hence the other atoms."

"I'll read about it and get the real picture. Know a good book?"

Thomas scratched his chin. "If you can find a copy of 'The Days of Creation,' by Willy Ley, the first part of the book has a description of the Solar Phoenix."

"Well, good enough," said Dr. Hamilton. "But just bear one thing in mind. You think you've beaten the engineer. Your basic trouble is just that the engineer is you, too. He has your ability and your knowledge and your experience upon which to work. He is no fool, and you can take that as a back-handed compliment if you want to. He is just as capable an engineer as you are a physicist. He thinks in different channels, I will admit. But, Thomas, remember that his extra-channellar thinking is done with the same thinking equipment as yours is, and it is no less efficient because of being divergent from your own thought-track. Your battle was won too easily to be conclusive."

"What do you expect?"

"I wouldn't know. I'm no scientist in physics." Hamilton held up a hand as Thomas started to protest. "I use 'scientist' despite your dislike of the word

only because there is no term that describes both of the attributes of practical engineer and research physicist. Frankly, I'm hoping for an eventual coalition, but I fear not."

"Why view no-coalition with distaste?" demanded Thomas.

"Because both personalities offer much to the world, to science in general, and to the body that houses both of them."

"I heartily dislike all aspects of practical engineering," stated Thomas flatly. "To be everlastingly forced to retrace your own steps, again and again and again, working out the most insignificant details—bah!"

"The engineer has another viewpoint."

"I know. But the engineer in this case is here only because of his own necessity—which he himself has removed. I am the real entity; I am the desire of the engineer. I am what he wants to be. *I am what he will become!*"

"Good morning, Frank."

"Morning, Miss Elaine. Mr. Lionel isn't here."

"He'll be back?" asked the girl.

"Oh yes. He went over to see Dr. Hamilton."

"Oh. Frank, the usual question?"

"This morning he is Thomas Lionel, Ph.D., M.M."

"Oh."

"He went to bed Tom Lionel, Consulting Engineer."

"I wonder if he remembers," smiled Elaine.

The *Poltergeist* landed on the lawn. It was silent, but a flash of sunshine caught the sleek side and attracted Elaine's attention.

"Hi," she called as she emerged from the house.

"Howdy," he answered. "What brings you out?"

"Never ask a girl a question like that," she laughed. "You'll never get the right answer."

"Why?"

"If she says 'you' it's either a lie or she's the kind of girl your mother tried to protect you from. If she says anything else, it's either a lie or she's the kind of girl your mother tried to protect you from."

"A man can't win," snorted Thomas.

"Does a man really want to win?"

"Nope," admitted Thomas. "I won't ask questions, Elaine. I'll just be glad you came."

"I'm glad you're glad."

Elaine flirted with him shamelessly, and then turned toward the laboratory building. He followed, and they kept up a running fire of light talk all the way.

"The first thing I have to do is to see what the engineer was doing last," remarked Thomas as he opened the laboratory door.

"You are a strange fellow," smiled Elaine. "You respect each other's possessions and beliefs, though you argue madly through impersonal mediums. Still writing nasty letters?"

"Uh-huh. And playing chess."

"What's he been doing?" asked Elaine innocently.

"Don't really know. Aside from some experiments on the poltergeist effect—reducing them to practice—I wouldn't know. I doubt that he's been doing much else. I do happen to know that he's deeply interested in the epicenter effect. He may find the key to it, too."

The laboratory was about as he remembered it. There were some changes. A few of the pieces of equipment were moved; some of them were converted; and a couple of them had been built in to other, larger pieces. All of the workmanship was clean and shining.

The cyclospectrograph had been worked on with a vengeance. It had lost its haywire appearance. The D plates were all neatly machined and the high frequency plumbing was all rearranged into mathematical and technical symmetry. The hours-use counter showed constant operation for several days solid, which interested the physicist.



"He's found a use for it," he grinned at Elaine.

"He finds a use for most everything," she said. "He's a pretty sharp man."

"Thanks," grinned Thomas, recalling what the psychiatrist had said regarding the mutual efficiency of the mind in Thomas Lionel's body.

"Wonder what this crystal is," muttered Thomas.

"Looks like a natural quartz."

"Might be—though I doubt it."

"Can you find out?"

"Eventually. If it is interesting, I will. What bothers me mostly, though," said Thomas thoughtfully, "are two things. One of them is that open drum of gooey tar. The other one is that vat of used motor oil."

"The oil I understand. But what is the tarlike goo?"

"I forget its name. It is one of the natural asphalt family and it ranks high—along with chewing gum—among those substances in which I would least like to bathe."

"Um. I detect a tone of distaste," laughed Elaine. "Here's another little tricky gadget. Looks sort of like your tungsten box."

"Oh?" asked Thomas.

"Yes ... say, Tommy, what's an epicenter?"

"Ah ... why?" asked the physicist, his attention on the cyclospectrograph.

"This box has a little sign on it. It says: 'Be an epicenter' and some other stuff."

"The epicenter is the main feature around which the phenomena revolves," explained Thomas idly.

"Oh."

Elaine fondled the little box. Her forefinger touched the button, felt its smoothness. In her mind was knowledge of the dire effects caused by tyros who push strange buttons. Certainly there was no curiosity deep enough to override her own good sense. But subconsciously the natural impulse to

touch wet paint, to kick the package on the sidewalk, came to the fore and Elaine stood there, looking the box over with her forefinger set against the button.

"Be an epicenter'," she repeated.

It registered. Like a swift montage, events past, present and future sped through Thomas Lionel's mind. He went from the basic idea to the foregone conclusion in three lightning-quick steps.

"NO!" he yelled.

But it was too late.

And through his mind there passed a vision that made him swallow. Elaine—dressed in a simple frock of printed silk, garnished from the top of her beautifully coiffed hair to the bottoms of her exquisitely shod feet in an awful mixture of used crankcase oil and a tar-asphaltum—

In vain he tried to cross the twenty feet that separated him from the girl. In vain he tried to get there, to snatch that devilish box from her hand, to grab it and hurl it far enough away so that the effect wouldn't even cause a bad splash.

The idea of seeing her all gooed up. That made him shout hoarsely.

It shouldn't happen to a dog—

And then it hit him. He was fully fifteen feet from the girl and her little instrument. A half-hour's observational time went into milliseconds in Thomas Lionel's mind as he watched the open drum of asphaltum compound rise out of the open top in a parabolic arc. It arched high, just missing the ceiling, and passing in an ogee curve to miss a stanchion. Forward it came, to curve downward upon his own bare head.

Simultaneously, he was drenched from behind by the arching column of oil from the vat behind him.

In twin, converging arches, Thomas was inundated and thoroughly soaked from head to toe with a whirling mixture of oil and tar.

He cleared his eyes with squeegeeing fingers. Elaine, holding in her laughter with effort, showed him the box.

Above the button it said:

BE AN EPICENTER!

Control that mysterious power. Exert the
forces of hidden nature in your behalf!

PRESS HERE!

"He's found it," croaked Thomas. "He's found it!"

"And you're a mess."

"That I am," said Thomas shaking off some of the gluck that was trickling down his arm. "That I am."

"Don't you mind?"

"I have my own revenge. My own, particular means of revenge. I'm sorry, Elaine. I must now leave you. The engineer has had his fun—now, my sweet, he may have the compensating task of cleaning up!"

Thomas turned and found the little tungsten box with its label: BE AN ENGINEER! and pressed the button.

Tom Lionel, Consulting Engineer, removed his finger from the button, and turned to see Elaine.

"Was it funny?" he asked.

"Very much so," she laughed.

"Who pushed it?"

"I did."

"Too bad. I'd rather he got it by his own machinations."

"He tried to stop me—"

"Uh-huh. Maybe it's better the way it is," Tom laughed in spite of the load of uncomfot he was carrying. He wiped some of the oil and tar mixture from his face and continued. "The instantaneous feeling of horror at the

idea of seeing you glucked over with this mess must have given him some shock. No doubt he thought that whatever would happen would happen to the holder of the epicenter locator."

"Now what are you going to do?"

"Me?"

"Yes. You're going to clean up, aren't you?"

"Not me."

"How are you going to ... to—?"

"Cause his return?"

"Yes."

Tom considered. "I guess I'm licked. He'll just use this box of his."

"Can't you undo it?"

"Nope. It's just too tough. I'd go to work on the insides with acid if I could get inside of it. The outside is possible, but I haven't enough acid to react with the whole darned box. But I'm going to get something. Well, I'm going inside and take myself a shower. Wait—I'll be back."

An hour later, Tom Lionel emerged from the bathroom. Frank, the houseboy went in with a humorous shake of the head. He'd seen the embryonic mess and knew what there was to do.

"Now what?" asked Elaine.

"Well, you see, the thing is slightly out of hand," exclaimed Tom. "I started this thing because my physicist friend got out of line and shot the entire bankroll on a pile of scientific flapdoodle." He took a cigarette case from his pocket that glinted and iridesced as he opened it. "I've been able to use nearly everything," he grinned, "including the ruling engine," he waved the grating-ruled cigarette case at the girl. "Marten shelled out about ten thousand bucks for the secret of the finish on this case. He's ruling jewelry now and it is the largest thing since the discovery of diamond-faceting. I'm

also getting a five percent royalty on every grating-ruled piece that's made. It ain't hay.

"Anyway, it backfired on me because I presented him with something that offered him, not frustration, but instead, he proceeded to make something of it that no sensible engineer could ignore. And," he continued ruefully, "it did two more drastic things. One, it made his continued influence necessary. There are too many things that he knows to dispense with his type of thinking. Number two, my success in reducing his discoveries to practice has resulted in the generation of a good income. That has been the basis of our argument. He's impractical to the extreme, but as long as the body is fed, both materially and intellectually, so what? So instead of finding myself the winner, I'm actually fighting for my own existence." Tom went bitter. "A fine thing. To be forced to fight for one's existence because of factors that emanate from his own success."

Elaine put a hand on his shoulder. "Don't be bitter," she said softly. "I ... I'll miss you—"

"Oh, don't worry," he told her in a strained voice. "I don't intend to give up." He cradled her face between his hands and looked her straight in the eyes. "If, as, and when, I—though the concept is purely hypothetical—might possibly lose—mind, I have no intention of losing since I intend to win unconditionally and maintaining the present status is intolerable—the other guy will have been in such a mad battle that he'll be forced into accepting some practical tenets as a factor. Then he'll be more like me."

"This may hurt," she said seriously, "but you are not as different as you might think."

"He hates the thought of practicality."

"And yet," said Elaine, "if all were engineers who would take time to seek out the little-known facts?"

"And," snorted Tom, "if all were physicists, we would still be hanging from trees, tossing coconuts at one another whilst a few bright dawn-men were contemplating the possibilities of using fire—but, of course, doing nothing about it. After all, once the physicist has considered all the angles, he's through. He doesn't give a howling hoot whether what he's considered is practical—after all, it *is* interesting and that's all he cares about."

"But—"

"Since the physicist's thinking is actually based upon past proof—made by practical engineers—the contemplation of fire would be as far as they'd get. For there would be no engineer to ever use it to show its practical possibilities! That's based on my horribly hypothetical world where all were physicists and none were engineers."

"Who invented the bow and arrow?" asked Elaine.

"Ab, Ug, or Unguh. He, she, or it was an experimenter. Y'see, Elaine, at one time there were neither engineers nor physicists. Alexander Graham Bell was not—in our present day sense—a physicist nor was Morse, or Edison or Lodge. Nor were they engineers. Somewhere since then the line has been divided. In them days they were basement geniuses. But now," he said bitterly, "there is one set of people who think up cockeyed things and another set that figures out what to do with them."

"Y'know," smiled Elaine, "I think that getting together would be the finest thing that ever happened to you and the physicist."

Tom backed up three steps. "Look," he snapped, "I've heard a lot. I can stand for a lot. But that's something that I can't even consider."

"Both of you offer so much to—"

"Yeah," he sneered, "and we're both solid citizens! Hooey."

Tom stalked over to the chessboard and looked down. "Overconfidence is a dangerous thing," he said with a smile. He moved a bishop halfway across the board. "There," he said with a satisfied air, "that should be obscure enough to fool anybody, even Lasker."

"Who's Lasker?" asked the girl.

"One of the chess masters."

"Oh."

"Now," he said, "I'm going to ask you a favor."

"Yes."

He grinned. "It might be quite personal."

"In which case I'd ask you a favor, too."

"What kind?"

"That depends on the nature of the original request. What's yours?"

"I'd like you to write a few letters for me."

"In which case there'll be no counterproposition."

For three solid hours, Elaine sat at the typewriter. At the end of that time, Tom smiled, patted her on the top of the head and said: "You've been a good girl, kiddo."

"Thanks," she mumbled. "But there are things about your physicist that I do admire. He never makes his women work."

"Impractical lad," laughed Tom.

"Impractical, but fun."

"Bah. You, too, huh?"

"Well fun is—"

"Impractical."

"When you find time to be impractical," said Elaine, rising, "you may invite me over long enough to find out just exactly how practical an impractical batch of fun can be. Practically, I am an impractical asset with indispensable attributes." She arched one eyebrow at him and leered in a ladylike fashion. "You'll find out," she told him.

"Dinner?" he suggested.

"That I cook? Ah-ahhh." She touched his cheek lightly and then said: "I gotta go. I'm late as it is. Sorry, Tommy. But that's how it stands. Take it easy—and I'll be seein' you."

Tom Lionel, Consulting Engineer, saw her away, and then returned to his desk full of work. He sorted papers, did some computations, manipulated some theory, and then sat still, thinking out his plan.

His evening was full. He experimented in the laboratory until the wee small hours, and then spent another two hours contemplating, with relish, the results. He finished by writing another letter, taking a last look at the chessboard, and then retired with a final look at the calendar.

Thomas Lionel, Ph.D., M.M., awoke with a feeling of self-satisfaction. The world was his onion and he knew it. There was nothing to detract from his success. After all, every time he returned it was because the engineer had been frustrated. The same thing had happened again.

He breakfasted lazily, reading the mail and the notes made by the engineer. The notes satisfied him. He added some notations and made some calculations himself that would further frustrate the engineer when again possession of the body changed minds. He noted with relish that the bank account was growing by leaps and bounds—a backfired result of the engineer's own machinations.



How long it would be before the engineer was completely vanished he did not know, but it would not be too long.

A few more developments of the poltergeist effect, another series of new sciences—with their attendant publicity, sales, and, of course, royalty and licensing fees—and the engineer would find nothing in his life worth living for. He then would turn, bending his naturally curious mind to the more obscure realm of physics.

In other words, the mind of Tom Lionel would become congruent with the mind of Thomas Lionel. Tom Lionel would vanish. Thomas Lionel, the ultimate desire of the engineer's mind would take control and the period of schizophrenia would end.

It was more than just logical. The consulting engineer wanted to be a physicist. Now that all barriers were removed, he would.

The hearty dislike of physicists that characterized the mind of the engineer was sheer jealousy; psychological block; that factor of the mind which, when denied a desire, hates all others who successfully achieve it.

Aesop called it "Sour Grapes."

He noted the calendar for the day. He nodded. He was to see a group of physicists from one of the government bureaus. That would be O.K. Later in the afternoon there was a conference between a group of production superintendents who were about to start producing items using the poltergeist effect. He made a notation to investigate the epicenter effect and see what could be generated out of it.

The latter was interesting, and presented a problem.

He arose from the breakfast table and sought the chessboard. He laughed quietly and advanced a knight to cover his opponent's bishop. That was in accordance with a well-known theory of chess. If, after several moves, no apparent pattern is manifest, attack with a minor piece—or even capture or exchange minor pieces. The plan of attack may be obscure to you, but the chances are that a bold counterattack or exchange of minor pieces will disrupt the attack.

Well, all was well.

His plans? They were simple enough. He would carry on. He would do more research, and confound the engineer whenever next he returned. It was as simple as that.

"The poltergeist effect," said Thomas, facing the group of government physicists, "is still in the field of research and development—and wide open on all sides. Much is still unknown about it, in spite of the fact," he added distastefully, "that certain aspects of it have already been put to work.

"You are all familiar with the initial theories, though I shall touch upon them briefly here again. The spatial strains, when under the influence of

solar gravity, lunar gravity, and terrestrial gravity all in proper vectorial relationship, add to the spatial strains caused by the magnetic fields of the three main bodies mentioned. The electrostatic fields in the terrestrial atmosphere—the same which cause lightning in another manifestation—also add to the spatial strain. These are all natural phenomena.

"The radiation of a thinking brain, gentlemen, is not a natural phenomena—not a natural characteristic warp in space," he corrected with a slight smile. "It causes a vibration in the region of the end-hysteresis loop of space itself when space is saturated to that threshold by the natural warps and strains. It exerts a triggering action that releases a more powerful nonphysical radiation, which in turn releases the space strain that causes—things—to move. Also this final buckling of the space strain releases a component of the initial wave which again exerts the triggering action. The proposition is self-sustaining as long as the gravitic, magnetic, and electrostatic effects are such as to maintain the saturation of space at the hysteresis-loop level.

"However," continued Thomas, "it has been discovered that these potentials are not only released, but are dirigible, under the complex force-fields generated by the equipment so far developed. Some effects are akin to gravitic effects—the real nature of which none of us can state, or even guess."

"Your statements border on the fantastic," smiled the head visitor, Lowell Johnstown. "Were it not for the fact that your statements have been accompanied by experimental proof, we would be forced to discount them as the ravings of an insane mind. However, your paper before the American Physical Society plus your experimental data—which we all have duplicated—gives proof. The nullification of gravity—"

"Not nullification," insisted Thomas. "I do not claim nullification. The effect is a development of diagravitic force."

"The difference—?"

"Does exist. The generation of a counter-force may, in some cases be considered nullification. But nullification does not describe all aspects of counter-force. I prefer to use the counter-force definition, since vectorial components may be generated in the object under observation. These forces have no relation to the force of gravity acting upon the object, other than

can be similarly observed in the free flight of a projectile, where both the forward motion and the gravitic attraction cause summation of forces into a parabolic path."

"I'll accept your remarks. But we are here to discuss the epicenter effect."

"The epicenter is a generator of the radiation which causes release of the potentials indigenous to all material bodies. This radiation is of a complex nature. It requires both physical, electrostatic, electromagnetic generators to produce the radiation that triggers the poltergeist potential.

"Incidentally, gentlemen, this may be why some people always seem to be getting into accidents. I believe that we have the true answer to the 'Accident-Prone' within our grasp."

"How does this manifest in experimental work?" asked Johnstown.

"By crystallographic generation of the force-fields coupled with the radiation of the brain in question. The effect is probably more pronounced with a man whose prevalence toward poltergeist effects and accident proneness is higher than normal. I am seeking such a man now. By further filtering through crystals the random release of poltergeist energy is directed and controlled so that a desired object may be moved almost at will."

"This equipment—can we hope to get it in practical size?"

Thomas considered and then said with a touch of disinterest: "It is possible. No doubt the practical viewpoint will appeal to many."

"To sum up your statements," interjected Johnstown, "we have an effect that will cause the movement of any desired object by the use of nonmaterial supporting and actuating means."

"Right. And one other interesting effect. The generation of direct current electricity is also possible in the same manner. I might even add," he concluded with a smile, "that manifestation of any physical effect is possible."

Johnstown packed the pages of data in his briefcase. "You have unlocked a veritable universe of basic study," he said. "You should feel gratified. We'll keep in touch with you, Lionel. And, we'll return once we have had a

chance to digest this information. Also, we'll furnish you with whatever observations we make."

Thomas watched them leave. He smiled. Adding to the discomfiture of the engineer at really having nothing left to work for—the bank account being filled daily—he was tossing his discoveries to the world of physicists, and other engineers and scientists would take over, more than likely leaving the engineer foundering in a sea of uncertainty.

The group of production superintendents entered and seated themselves. Their spokesman, Charles Norden said: "We are here because of certain difficulties we are having in making your effect operate satisfactorily."

"I can, of course, assist you," smiled Thomas, affably.

"Good," answered Norden. "Here is the first difficulty." He pulled from his case a sheaf of blueprints, and he spread them out across the desk. "Our design department claims that the arrangement of knobs on the panel is inconvenient. A suggested change is to put them like this."

Franklen, who was one of Norden's associates objected: "That means you'll have to shunt the snivvy over here. That lengthens the leads and cause instability."

"But we can reinstate the stability by running the leads through a dingink."

"That won't help. Shielding the leads only adds distributed capacity."

"It works. Only one percent loss in efficiency and better stability, believe it or not."

"Well," said Norden, "I'll leave it up to Lionel, here. What do you think?"

"I'll have to consider it," answered Lionel.

"You understand that it is important," urged Norden.

"But why?"

"Why?" exploded Norden. "Great Scott! Look Lionel, the arrangements of these knobs are such that the operator must cross the calibration-scale with

his hand while adjusting the output. That means that he must either assume a cramped position or he must adjust, observe, adjust, observe, and so on, taking the adjustment of output by increments instead of a stepless arrival at the precise value."

"Um."

"Look, Lionel, we aren't toying with the job of lifting a standard weight. We are hoisting three hundred tons of semirigid structure that mustn't be juggled too much."

"I see. Well under the circumstances I'll take quick action and give you the answer within twenty-four hours."

"We can't have it immediately?"

"I'm afraid not. I must make some tests before I can pass judgment on the matter."

"We are more or less forced to accept your time-limit," said Norden glumly. "Please understand that time is essential because every minute that our production line is down costs the company about twenty dollars."

"I'll get your answer in twenty-four hours," Thomas faced another man. "And your trouble?"

He was Mawson, of Technical Manufacturing, Inc.

"You specify this part," indicating another set of blueprints, "as pure copper. Anything else do?"

"It carries high frequency. Copper is best—unless you could get silver. If that is—"

"Look we're making production and hope to hit fifteen thousand completed assemblies per day. That piece would weigh about two pounds. Silver is out."

"What's the matter with copper?" asked Lionel.

"It cuts like cheddar cheese, gums up the tools, and is generally not good for close tolerance work."

"The first one was all right."

"Listen," said Mawson, "you carved the first one out by hand and I'll bet it took you four or five hours. We're going to 'run 'em on an automatic screw machine at the rate of ten per minute."

"So?"

"I want to use free cutting brass."

"You'll lose conductivity—"

"The rig will be only seven percent less efficient. Tests—"

"Your tests may be right. But seven percent loss is pretty bad," grumbled Thomas. "I'd say no."

"Then I'm going to ask you to name a substitute. What alloy would suffice? I want a free cutting alloy that'll come off of the tools clean."

Will White spoke up at this point. "We've got a bit of regeneration in our driver system," he said.

"That's easy. Have one of your engineers remove it."

"Can't. Anything they do to remove regeneration also destroys the driver's efficiency."

"Efficiency," said Thomas, "is the inverse function of the frequency of drive divided by the number of full-phase poles in the genedyne. Expressed by the quadratic equation in which A equals the number of full-phase poles—"

"But we can't get rid of the regeneration!" exploded White.

"Look, that driver is as simple as ABC. It has and will be a standard assembly for seven generations past and forward. Now don't tell me—"

"I'm telling you that we're burning up our test stands left and right. How long can you take a forty percent regeneration in a hundred kilowatt genedyne?"

"Well, not long," admitted Thomas. "You understand the principles involved?"

"Not entirely."

"I'll explain. The force-fields created by the full-phase poles under the power output from the driver create a nonelectromagnetic field radiation. The intensity of this field is a function of the driver output, derived tertially through the pole system. Now the development of the field radiation creates a space hysteresis that—"

"Look," snapped White, "I don't give a care about field theory. It doesn't apply in my job. I merely want to know what to do about the regeneration."

"How can you work without understanding the theory?" asked Thomas with a sneer.

"How can a musician play a pipe organ without a course in first year physics?" shouted White. "Just tell me what to do!"

"Tell 'em to retune the driver to another band."

"They aren't certain that the pick-up loop is itself tuned or not."

"It is."

"Then that means tuning the entire feeder line."

"Naturally."

"But the feeder line is a silver plated die casting."

"Change the dies," said Thomas.

"May we charge you for them?" asked White, sweetly. "And also for scrapping the three hundred and seventy thousand parts we have cast already?"

"Why get so far ahead?"

"Look, Thomas Lionel, we did it because it was cheaper not to tie up the die-caster's shop in weekly driblets rather than get the whole order in a lump. Now—what do we do about regeneration?"

"Anybody can clear up regeneration," snapped Lionel.

"Good—you show us how. That's what we're retaining you for. Your developments and whatever technical assistance is required."

Thomas Lionel looked up, and scanned each face at the big table. "From left to right, can you state your problems?" he asked.

"Plating specs on the genedyne."

"Problem in suspension during process."

"Can plus or minus fifteen percent electrical components be used in place of ten percents?"

"We still require the alignment procedure."

Thomas held up a hand. "O.K., fellows. Submit your problems in writing and I'll furnish the answers in twenty-four hours."

Norden bobbed his head in agreement. "I fear that I shall call for cancellation of contract and the forfeit sum if your answers are not forthcoming."

Thomas nodded silently. The forfeit—if this whole gang bopped their contracts back he'd lose his shirt.

He watched them file out. And his eyes dropped from their stare out of the window to the pile of questions on the desk. A pile of production problems!

Sheer, unadulterated hell.

Well, he might as well call the engineer and let him handle this. It was one of the things that the engineer got a kick out of. As a physicist, this was not his job—and as an intelligent physicist, he did know how to get things done. Everything for its own use; if he didn't know, he knew where to find out.

Thomas went out to the laboratory and faced his tungsten box. A twinge of wonder flashed through his mind. It bothered him.

Was this an admission of partial defeat?

Not at all. This was good sense. Call in the engineer to clear up this mess, since his income and well-being depended upon it. Once these contracts were clear and closed—well, time enough!

He pressed the button.

And Thomas Lionel, Ph.D., M.M., pressed it again.

And again—

And again.

And then took a suspicious glance at the oxy-hydrogen torch on the table. A growing fear hit him. Tungsten wouldn't budge under an acetylene torch. Acids were not too effective, and plain, old-fashioned cutting tools were sheer foolishness. But heat the block white hot and hit it with an oxygen lance—

Thomas looked on the back side.

Uh-huh. The engineer had enjoyed himself. The back side of his little tungsten wave generator had been poked full of ragged holes; cut in ribbons with the oxygen lance, and generally made messy. The wave-guides and channels were all un-terminated and laid open. Pushing the button wouldn't do a thing.

It definitely would not call the engineer.

He had twenty-four hours to solve the production people's problems.

And Thomas Lionel understood. The engineer had his own little trap. No doubt the engineer would go fishing if called, and only the physicist was really interested in fighting this thing out to the bitter end. The engineer, losing already, had only a bank account to throw away by not working. And the engineer could get another one soon enough if permitted to do so.

Twenty-four hours.

Thomas headed in toward the house to get his notebook and his engineering notes. He'd have to take over the engineer's job, no matter how distasteful.

Out of one corner of his eye he saw the chessboard and he stopped short. It had been the odd angle that gave the trap away but his moving of the knight had opened a line right down into his own defenses. On the next move there would be a severe attack against his queen, and in saving that, he'd lose the bishop. If he sacrificed the queen, he was as good as lost. In fact, it was checkmate no matter how long he fought; no matter what he did, it was only a question of three moves minimum or seven moves maximum.

Well, might as well give up. This game was gone and there was no use in fighting a losing battle—whipping a dead cat—

He opened the drawer and removed the little sign normally used to terminate a game without going through the formality of a checkmate.

But the sign had been augmented. It read:



THE END.

*** END OF THE PROJECT GUTENBERG EBOOK TROUBLE TIMES
TWO ***

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